Project Proposal: Latent Evaluation of Reasoning Traces through Circuit Stability

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Abstract

Reasoning in language models remains confined mainly to mimicking human cognitive processes. The decomposition of complex tasks often depends on the model's heuristic tendencies, which may be influenced by memorized patterns rather than true generalization. We hypothesize that each model's internally learned behavior—reflected in its emergent circuitry—can be harnessed to improve reasoning and efficiency. In this work, we introduce a novel metric that leverages a model's intrinsic activation pathways for planning and task decomposition. This approach incorporates circuit-level insights into the reasoning pipeline, enabling more robust and adaptive problem-solving and representing a shift toward reasoning strategies that are explicitly grounded in how a model thinks.

1 Introduction

In recent years, many papers have tried to improve LLM reasoning capabilities. Since Chain-of-Thought [Wei et al., 2022] pioneered k-shot prompting, numerous works have since been built upon it, such as Automatic Chain-of-Thought [Zhang et al., 2023], Tree-of-thoughts [Yao et al., 2023], Buffer of Thoughts [Yang et al., 2024], and Forest-of-thoughts [Bi et al., 2024].

Existing work is heavily inspired by human cognition. For example, Tree-of-Thoughts draws from thinking observed in both humans and animals. System 1, which is fast and automatic, and System Two, which is slow and deliberate [Yao et al., 2023, Kahneman, 2011]. Chain-of-Thought is known to be similar to how humans approach problem-solving [Zhang et al., 2023]. However, research has shown that while a model can give convincing post hoc explanations, those explanations do not reflect the model's "thoughts" [Bubeck et al., 2023]. Oftentimes, when the next step is unclear, models hallucinate [McKenna et al., 2023] or fall back on memorization. So, relying solely on the model's heuristic reasoning does not always yield optimal results.

At the same time, our understanding of a model's internal behavior has improved significantly. Circuits can be interpreted as learned algorithms specialized for a specific class of tasks. For example, recall uses the induction heads, no matter the input length or lexical variance [Olsson et al., 2022]. Through automated circuit discovery [Conmy et al., 2023] and work on transformer circuits[Elhage et al., 2021], researchers have found natural circuits that exist in an LM during training and continue to exist as the model scales and learns.

Taking all this into account, we propose a metric that leverages our understanding of model internals to evaluate reasoning traces and solve complex problems.

1.1 Motivation

Despite significant advancements in large language models (LLMs), particularly through techniques like chain-of-thought (CoT) prompting and k-shot learning, and more recently, tree-of-thought

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(ToT) along with verifiers, these models still exhibit notable limitations. A prevalent issue is their susceptibility to errors, predominantly symbolic mapping errors and wrong/missing-step errors, as highlighted in previous research [Wei et al., 2022].

We hypothesize that these errors stem from a fundamental challenge: LMs may not possess a true semantic understanding of problems and often struggle to generate robust, unique problem-solving plans. Crucially, coming up with an effective plan is fundamentally the challenge of decomposing a problem in a generalizable manner. Current training methodologies, which rely on human-intuited steps, might not be optimally efficient for computational models. This could lead to misinterpretations of relationships between numerical and linguistic elements, or the misapplication of learned patterns from seemingly similar but distinct problems.

The persistence of these limitations is critical because it directly impacts the reliability of LMs and hinders their scalability. The extensive data requirements, demanding a diverse distribution of question types, exemplar reasoning steps, and accurate answers, become increasingly challenging to meet given the current error rates.

Our proposed solution addresses this limitation by introducing a novel approach to plan generation, leveraging "circuit stability" as a metric to devise "computer-natural" plans. This involves evaluating the quality of a plan *before* execution, moving beyond pattern reliance derived from training data. Conceptually, this extends process reward models to "plan reward models", where models are rewarded for generating plans that demonstrate higher circuit stability.

We anticipate that this approach will be effective due to the demonstrated success of process reward models in improving multi-step reasoning. By focusing on enhancing problem decomposition and planning—a critical pre-execution phase—we aim to significantly boost accuracy. Intuitively, plans that are highly sensitive to minor perturbations are likely less generalizable and may merely reflect memorized training patterns. Our method seeks to identify and promote plans that are more robust and truly reflect a deeper understanding of the problem structure.

1.2 Related Literature

1. COT: Improving LLM reasoning with the use of small, intermediate steps or 'plans' together with k-shot prompting [Wei et al., 2022]
2. TOT: Improving decision making through the use of multiple COT/ LLM generated chains and the use of self evaluating 'voting' to decide the course of action [Yao et al., 2023].
3. FOT: Improving efficiency and accuracy through the use of multiple tree of thoughts, input augmentation, and real-time correction and self-evaluation [Bi et al., 2024].
4. BOT: Improves reasoning through iteratively improving the initial prompt; the prompt consists of appending 'potential but not explored' paths as experience and the scoring is done through evaluating the node and the transition score [Chen et al., 2024].
5. Automatic Chain-of-Thought prompting reduces manual effort by clustering questions and using zero-shot CoT such as "think step by step" to generate diverse example chains for few-shot prompts [Zhang et al., 2023].

1.3 Contributions

1. We extend our understanding of circuit stability to evaluate plans generated by LLMs to find optimal ways to solve complex tasks
2. We attempt to make circuit stability scalable to plans and reasoning steps easily for prompting techniques such as TOT, to make it easier for circuit stability to be adopted more widely, and to highlight the improvements brought by circuit stability by testing it on multi-step, complex reasoning problems.
3. We also plan to bring about new evaluation metrics that focus more on the creation of robust plans and reasoning steps for each plan by LLMs instead of just focusing on output values (which is less interpretable and more prone to memorization).
4. Our research will guide LLMs towards being more interpretable and better at solving complex, multi-step problems, leveraging circuit stability, which would become a convenient standard for evaluating the generalization nature of LLMs to new problems, replacing the use of datasets that are hard to create and easily get saturated.

5. We believe that circuit stability can not only act as a path to better reasoning on problems during inference, but it can also become the new standard to test and/or train models. Models with better circuit stability could be thought of as better reasoning models, as they adapt to problems of similar tasks with perturbed or increasing complexity in inputs. Further research could lead to scalable methods for training LLMs to maintain circuit stability for better outcomes.

2 Methods

Research on automatic circuit discovery and internal circuits has established the idea that each edge can be treated as a binary variable, where edge $\in \{0, 1\}$. Using this representation, one can identify a computational subgraph [Ferrando and Voita, 2024, Conmy et al., 2023]. Alternatively, the work of Sun [2025] on circuit stability illustrates the concept of *soft circuits*, in which each edge is assigned an importance score drawn from a continuous and relaxed set, where edge $\in [0, 1]$. This aims to provide richer structural insight into the internal computations of the model.

Our metric for evaluating a step is heavily grounded in the work of soft circuits, while the rest of our method is very similar to existing research. We see our metric being integrated into other research without major changes in the workflow. A specific instantiation of our method follows two steps:

1. **Thought generation:** Given a complex task T belonging to a particular class of problems, we assume that it can be subdivided into a finite number of thoughts or subtasks:

   $$T = \{z_1, z_2, \ldots, z_i\}, \text{where each } z_i \text{ represents the } i\text{-th thought or subtask.}$$

   We use the *propose prompt* method from Yao et al. [2023] to generate a set of candidates for the first subtask z_1. We set the depth *m*, to be the number of variations for a particular thought z_i. So for each subtask z_i we have a set of variations *m*

   $$\mathcal{V}_i = \{z_{i1}, z_{i2}, \ldots, z_{im}\}, \quad \text{where each } z_{ij} \text{ denotes the } j\text{-th variation of subtask } i.$$

2. **Thought evaluation:** We take the set of *m* variations of the subtask z_i and compute a soft circuit for each variation z_{ij}. We call this matrix of continuous edge importance scores as the soft circuit matrix A_{ij}, where i and j represents the i-th subtask and j-th variation and not its dimensions.

   We plot each of the soft circuit matrices A_{ij} on a distribution graph to calculate two key variables: *skewness* and *similarity*. Based on the skewness, we take the most skewed activation matrix, which implies that it used the smallest subset of edges to compute the output. We interpret this as how finely the model has decomposed the subtask.

   We then score each candidate z_{ij} using the corresponding A_{ij}, by combining how sparse it is (skewness) and how different it is from the previous step (similarity), weighted by α and β. Then we pick the candidate with the best score. We define the scoring function as:

   $$\text{score}(z_{ij}) = \alpha \cdot \text{skewness}(A_{ij}) - \beta \cdot \text{similarity}(A_{ij}),$$

   and select the next step via:

   $$z^\star = \arg\max_{z_{ij} \in \mathcal{V}_i} \text{score}(z_{ij}).$$

3 Experimental Setup

We would use the model solve multi-step reasoning problems that benefit from planning, which we will generate through TOT and similar prompting techniques. For convenience, we will begin the experiment with mathematical problems because they are easier to judge the factual correctness of. The baseline would be responses and outputs generated just with TOT without circuit stability. We will be measuring the circuit stability of plans and the accuracy of models.

4 Benchmark and Evaluation

For the actual problems, we will use examples from datasets like GSM8K and PRM800K, as they are widely used and accepted for testing the reasoning ability of LLMs. We could plot the correlation between circuit stability and accuracy using statistical tests such as p-value, or give an estimate of how much circuit stability improved reasoning traces and outcomes over our experimental data. We would also show how circuit stability not only improves the final output, but also makes the model converge faster and more easily towards the ideal reasoning traces/plans that lead to it. We might give a penalty or a 0 score to situations where faulty reasoning led to correct outputs for both tests of simple TOT vs circuit stability, as we aim to focus on better reasoning abilities throughout the plan, instead of luckily getting the right answers with wrong assumptions.

5 Ideal Results

In the ideal results, better circuit stability will lead to better reasoning abilities and correct outputs in our model, and we would demonstrate that the same soft circuits can be used for solving another problem with slightly perturbed input prompts. We would also try to automate the process of generating and maintaining these soft circuits so the strategy to use circuit stability can take a step towards becoming model-agnostic and being applicable to more widespread practical uses in industry and academia.

6 Potential Limitations

The first step that determines the success of our project would be recreating the soft circuits for visualizing circuit stability for our experimental setup, and then being able to dynamically apply the same algorithm to several reasoning paths generated by the model for an input prompt. This could prove challenging due to the mathematics and coding involved in this process. The second challenge would be to score/evaluate the different plans generated by TOT and select the best one based on circuit stability consistently. Then, we would attempt to make this process task-agnostic and possibly model agnostic so it becomes more generalizable and easy to use in practical contexts. We cannot be confident about the strength of correlation between circuit stability with our TOT approach, but that's a reasonable uncertainty; we need experimentation before we can make any claims on whether we get the ideal results or not.

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